Import Library

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Load Data

```
data = pd.read_excel(r'D:\Daiict_Py\PY self study\swiggy data analysis\swiggy_data.xlsx')

data.head()
```

```
       State       City Order Date           Restaurant Name  \
0  Karnataka  Bengaluru 2025-06-29  Anand Sweets & Savouries
1  Karnataka  Bengaluru 2025-04-03     Srinidhi Sagar Deluxe
2  Karnataka  Bengaluru 2025-01-15     Srinidhi Sagar Deluxe
3  Karnataka  Bengaluru 2025-04-17     Srinidhi Sagar Deluxe
4  Karnataka  Bengaluru 2025-03-13     Srinidhi Sagar Deluxe

                Location     Category              Dish Name  Price
(INR)  \
0  Rajarajeshwari Nagar        Snack  Butter Murukku-200gm
133.9
1               Kengeri  Recommended            Badam Milk
52.0
2               Kengeri  Recommended        Chow Chow Bath
117.0
3               Kengeri  Recommended           Kesari Bath
65.0
4               Kengeri  Recommended            Mix Raitha
130.0

   Rating  Rating Count
0     4.0             0
1     4.5            25
2     4.7            48
3     4.6            65
4     4.0             0
```

Meta Data

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 197430 entries, 0 to 197429
Data columns (total 10 columns):
 #   Column           Non-Null Count   Dtype
---  ------           --------------   -----
 0   State            197430 non-null  object
 1   City             197430 non-null  object
```

```
 2   Order Date       197430 non-null  datetime64[ns]
 3   Restaurant Name  197430 non-null  object
 4   Location         197430 non-null  object
 5   Category         197430 non-null  object
 6   Dish Name        197430 non-null  object
 7   Price (INR)      197430 non-null  float64
 8   Rating           197430 non-null  float64
 9   Rating Count     197430 non-null  int64
dtypes: datetime64[ns](1), float64(2), int64(1), object(6)
memory usage: 15.1+ MB
```

```
data.shape
```

```
(197430, 10)
```

```
data.isna().sum()
```

```
State              0
City               0
Order Date         0
Restaurant Name    0
Location           0
Category           0
Dish Name          0
Price (INR)        0
Rating             0
Rating Count       0
dtype: int64
```

```
data.duplicated().sum()
```

```
np.int64(27)
```

```
round(data.duplicated().sum()*100/data.shape[0],2)
```

```
np.float64(0.01)
```

```
data.drop_duplicates(inplace=True)
```

```
data.shape
```

```
(197403, 10)
```

```
data.describe(include='all')
```

```
             State       City                         Order Date Restaurant
Name  \
count       197403     197403                             197403
197403
unique          28         28                                NaN
993
top     Karnataka  Bengaluru                                NaN
```

```
McDonald's
freq         20072      20072                            NaN
13528
mean           NaN        NaN  2025-05-01 19:42:28.028145152
NaN
min            NaN        NaN            2025-01-01 00:00:00
NaN
25%            NaN        NaN            2025-03-01 00:00:00
NaN
50%            NaN        NaN            2025-05-02 00:00:00
NaN
75%            NaN        NaN            2025-07-01 00:00:00
NaN
max            NaN        NaN            2025-08-31 00:00:00
NaN
std            NaN        NaN                            NaN
NaN

           Location     Category        Dish Name    Price (INR)  \
count        197403       197403           197403  197403.000000
unique          977         4972            59064            NaN
top     Gomti Nagar  Recommended  Choco Lava Cake            NaN
freq           2091        24098              303            NaN
mean            NaN          NaN              NaN     268.501413
min             NaN          NaN              NaN       0.950000
25%             NaN          NaN              NaN     139.000000
50%             NaN          NaN              NaN     229.000000
75%             NaN          NaN              NaN     329.000000
max             NaN          NaN              NaN    8000.000000
std             NaN          NaN              NaN     219.222980

               Rating   Rating Count
count   197403.000000  197403.000000
unique            NaN            NaN
top               NaN            NaN
freq              NaN            NaN
mean         4.341579      28.323637
min          1.500000       0.000000
25%          4.300000       0.000000
50%          4.400000       2.000000
75%          4.500000      15.000000
max          5.000000     999.000000
std          0.422585      87.547541
```

KPI'S

```
data.head(3)

       State       City Order Date           Restaurant Name  \
0  Karnataka  Bengaluru 2025-06-29  Anand Sweets & Savouries
```

```
1  Karnataka  Bengaluru 2025-04-03     Srinidhi Sagar Deluxe
2  Karnataka  Bengaluru 2025-01-15     Srinidhi Sagar Deluxe

                Location     Category              Dish Name  Price
(INR)  \
0  Rajarajeshwari Nagar        Snack  Butter Murukku-200gm
133.9
1               Kengeri  Recommended            Badam Milk
52.0
2               Kengeri  Recommended        Chow Chow Bath
117.0

   Rating  Rating Count
0     4.0             0
1     4.5            25
2     4.7            48
```

1. Total Sales

```
total_sales = data['Price (INR)'].sum()
print(f'total sales: {round(total_sales,2)}')

total sales: 53002984.47
```

1. Average Rating

```
avg_rating = data['Rating'].mean()
print(f'average rating: {round(avg_rating,1)}')

average rating: 4.3
```

1. Average Order Value

```
aov = data['Price (INR)'].mean()
print(f'average order value: {round(aov,2)}')

average order value: 268.5
```

1. Rating Count

```
ratings_count = data['Rating Count'].sum()
print(f'rating count: {round(ratings_count,2)}')

rating count: 5591171
```

1. Total Orders

```
total_order = data['Order Date'].count()
print(f'total orders: {total_order}')

total orders: 197403
```

Charts

```
data.head(2)
```

```
       State       City Order Date           Restaurant Name  \
0  Karnataka  Bengaluru 2025-06-29  Anand Sweets & Savouries
1  Karnataka  Bengaluru 2025-04-03     Srinidhi Sagar Deluxe

                Location     Category              Dish Name  Price
(INR)  \
0  Rajarajeshwari Nagar        Snack  Butter Murukku-200gm
133.9
1               Kengeri  Recommended            Badam Milk
52.0

   Rating  Rating Count
0     4.0             0
1     4.5            25
```

1. Monthly Sales Trend

```
from datetime import datetime
data.insert(3, 'Month', data['Order Date'].dt.month)

data['Month'].head(3)
```

```
0    6
1    4
2    1
Name: Month, dtype: int32
```

```
data.insert(4,'Year_Month', data['Order Date'].dt.year.astype(str)+
'-' + data['Order Date'].dt.month.astype(str))

data['Year_Month'].head(3)
```

```
0    2025-6
1    2025-4
2    2025-1
Name: Year_Month, dtype: object
```

```
data1 = data.groupby('Year_Month')['Price
(INR)'].sum().reset_index(name='Total Price')
data1
```

```
  Year_Month  Total Price
0     2025-1   6823981.33
1     2025-2   6268161.67
2     2025-3   6572738.20
3     2025-4   6590449.00
4     2025-5   6792621.35
5     2025-6   6513865.19
```

```
6     2025-7   6650268.51
7     2025-8   6790899.22

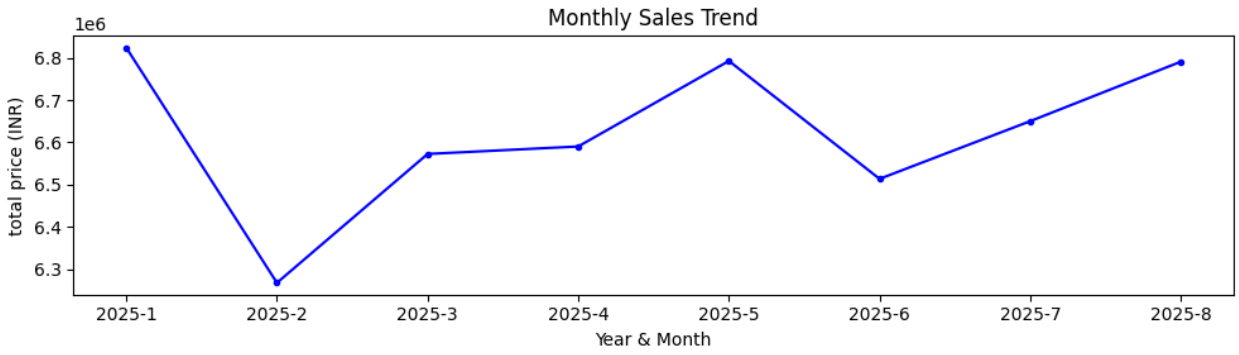
x = data1['Year_Month']
y = data1['Total Price']

plt.figure(figsize=(10,3))

plt.title('Monthly Sales Trend')
plt.plot(x,y,'b.-')
plt.xlabel('Year & Month')
plt.ylabel('total price (INR)')

plt.tight_layout()

plt.show()
```




```
data.head(2)
```

```
       State       City Order Date  Month Year_Month  \
0  Karnataka  Bengaluru 2025-06-29      6     2025-6
1  Karnataka  Bengaluru 2025-04-03      4     2025-4

            Restaurant Name              Location     Category  \
0  Anand Sweets & Savouries  Rajarajeshwari Nagar        Snack
1     Srinidhi Sagar Deluxe               Kengeri  Recommended

              Dish Name  Price (INR)  Rating  Rating Count
0  Butter Murukku-200gm        133.9     4.0             0
1            Badam Milk         52.0     4.5            25
```

1. Daily Sales Trend: Highliting order and revenue variations across days of the week.

```
data.insert(5,'Day_Name',data['Order Date'].dt.day_name())
data.head(4)
```

```
       State       City Order Date  Month Year_Month   Day_Name  \
0  Karnataka  Bengaluru 2025-06-29      6     2025-6     Sunday
1  Karnataka  Bengaluru 2025-04-03      4     2025-4   Thursday
```

```
2  Karnataka  Bengaluru 2025-01-15      1     2025-1  Wednesday
3  Karnataka  Bengaluru 2025-04-17      4     2025-4   Thursday

             Restaurant Name              Location     Category  \
0  Anand Sweets & Savouries  Rajarajeshwari Nagar        Snack
1     Srinidhi Sagar Deluxe               Kengeri  Recommended
2     Srinidhi Sagar Deluxe               Kengeri  Recommended
3     Srinidhi Sagar Deluxe               Kengeri  Recommended

               Dish Name  Price (INR)  Rating  Rating Count
0  Butter Murukku-200gm        133.9     4.0             0
1            Badam Milk         52.0     4.5            25
2        Chow Chow Bath        117.0     4.7            48
3           Kesari Bath         65.0     4.6            65
```

```
day_order = ['Monday', 'Tuesday', 'Wednesday', 'Thursday',
             'Friday', 'Saturday', 'Sunday']

data2 = data.groupby('Day_Name')['Price
(INR)'].sum().reset_index(name='Total Price')

data2['Day_Name'] = pd.Categorical(
    data2['Day_Name'],
    categories=day_order,
    ordered=True
)

data2 = data2.sort_values('Day_Name', ascending= True)

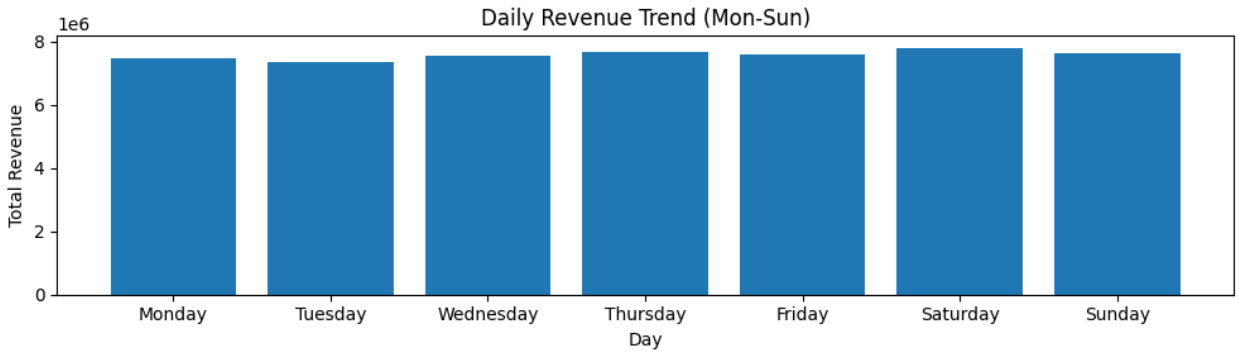
print(data2)

x = data2['Day_Name']
y = data2['Total Price']

plt.figure(figsize=(10, 3))
plt.title('Daily Revenue Trend (Mon-Sun)')
plt.bar(x, y)
plt.xlabel('Day')
plt.ylabel('Total Revenue')

plt.tight_layout()
plt.show()
```

```
    Day_Name  Total Price
1     Monday   7444475.14
5    Tuesday   7358996.01
6  Wednesday   7541692.52
4   Thursday   7663248.59
0     Friday   7579482.84
2   Saturday   7778260.40
3     Sunday   7636828.97
```



1. Total Sales By Food (Veg vs Non-Veg)

```
non_veg_keywords = ["chicken", "egg", "fish", "afghani", "mutton",
"prawn",
                    "biryani", "kabab", "kebab", "non-veg", "non veg"]

pattern = '|'.join(non_veg_keywords)

data['Food_Category'] = np.where(
    data['Dish Name'].str.lower().str.contains(pattern, na=False),
    "Non-Veg", "Veg"
)

name = data['Dish Name'].str.lower()
is_non_veg = name.str.contains(pattern, na=False)
is_veg_labeled = name.str.contains(r'\b(?:veg|paneer|soya|mushroom)\
b', na=False)

data['Food_Category'] = np.where(is_non_veg & ~is_veg_labeled, "Non-
Veg", "Veg")

data.head(3)
```

```
       State       City Order Date  Month Year_Month   Day_Name  \
0  Karnataka  Bengaluru 2025-06-29      6     2025-6     Sunday
1  Karnataka  Bengaluru 2025-04-03      4     2025-4   Thursday
2  Karnataka  Bengaluru 2025-01-15      1     2025-1  Wednesday

            Restaurant Name              Location     Category  \
0  Anand Sweets & Savouries  Rajarajeshwari Nagar        Snack
1     Srinidhi Sagar Deluxe               Kengeri  Recommended
2     Srinidhi Sagar Deluxe               Kengeri  Recommended

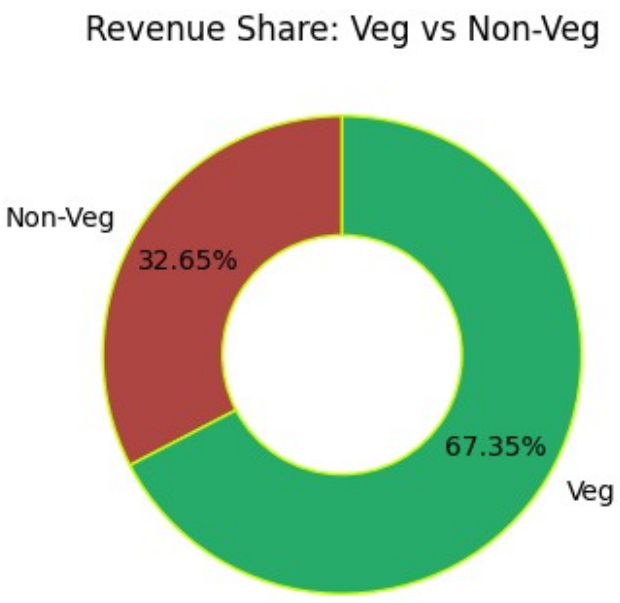
              Dish Name  Price (INR)  Rating  Rating Count
Food_Category
0  Butter Murukku-200gm        133.9     4.0             0
Veg
1            Badam Milk         52.0     4.5            25
Veg
```

```
2         Chow Chow Bath         117.0     4.7              48
Veg

food_revenue = data.groupby(['Food_Category'])['Price
(INR)'].sum().reset_index(name='Total Revenue')
food_revenue
```

```
  Food_Category  Total Revenue
0       Non-Veg    17304243.63
1           Veg    35698740.84
```

```
plt.figure(figsize=(4, 4))
plt.pie(
    food_revenue['Total Revenue'],
    labels=['Non-Veg', 'Veg'],
    autopct='%.2f%%',
    colors=['#AD4642','#26AB6A'],
    startangle=90,
    pctdistance=0.75,
    wedgeprops=dict(width=0.5, edgecolor='#C8FF00')
)
plt.title('Revenue Share: Veg vs Non-Veg')
plt.show()
```



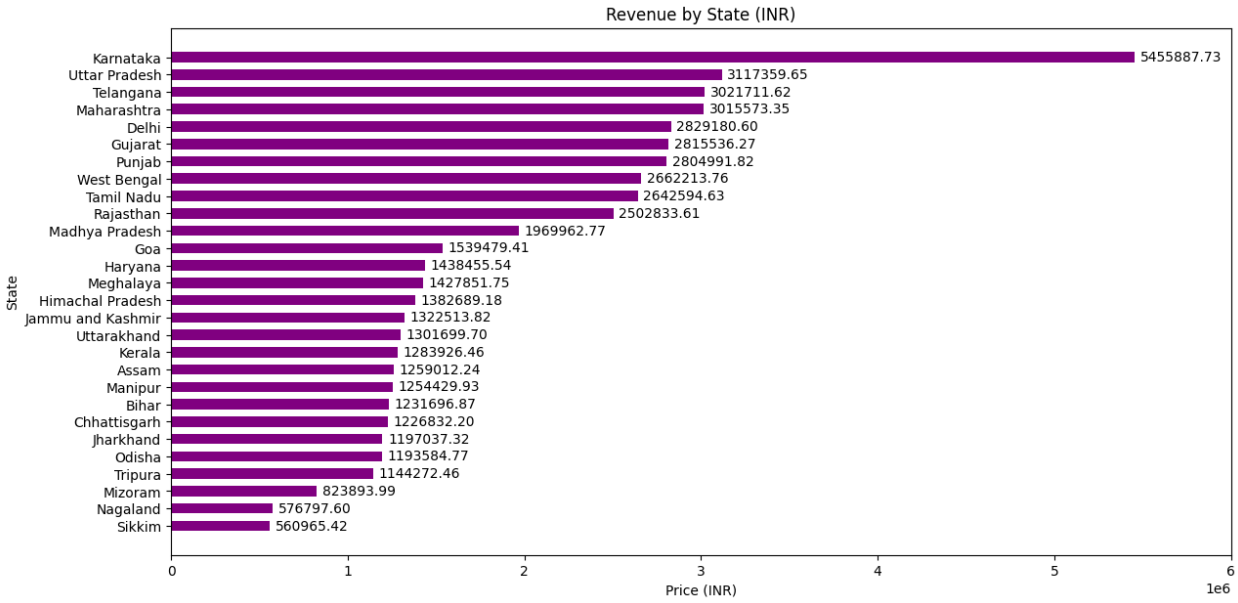
1. Total Sales by State

```
data['State'].nunique()
```

```
28
```

```
df = data.groupby('State')['Price (INR)'].sum().reset_index(name=
'Total_Revenue').sort_values("Total_Revenue")

plt.figure(figsize=(14, 7))
plt.title('Revenue by State (INR)')
bars = plt.barh(df['State'], df['Total_Revenue'], color='purple',
height=0.6)
plt.bar_label(bars, fmt='%.2f', padding=4)   # shows the full value
plt.xlabel('Price (INR)')
plt.ylabel('State')
plt.margins(x=0.1)   # extra space so labels don't get cut off
plt.show()
```



1. Quarterly Performance Summary- Combines sales, rating and order by quarter.

```
data['Order Date'] = pd.to_datetime(data['Order Date'])
data['Quarter'] = data['Order Date'].dt.to_period("Q").astype(str)

data.head(2)
```

```
       State       City Order Date  Month Year_Month  Day_Name  \
0  Karnataka  Bengaluru 2025-06-29      6     2025-6    Sunday
1  Karnataka  Bengaluru 2025-04-03      4     2025-4  Thursday

            Restaurant Name              Location     Category  \
0  Anand Sweets & Savouries  Rajarajeshwari Nagar        Snack
1     Srinidhi Sagar Deluxe               Kengeri  Recommended

              Dish Name  Price (INR)  Rating  Rating Count
Food_Category  \
0  Butter Murukku-200gm        133.9     4.0             0
Veg
```

```
1            Badam Milk         52.0     4.5             25
Veg

  Quarter
0  2025Q2
1  2025Q2
```

```
Quarterly_summary = data.groupby('Quarter').agg(
    Total_Sales = ('Price (INR)', 'sum'),
    Avg_Rating = ('Rating', 'mean'),
    Total_Orders = ('Order Date', 'count')
).sort_values('Quarter', ascending=True)

Quarterly_summary['Total_Sales'] =
Quarterly_summary['Total_Sales'].round(0)
Quarterly_summary['Avg_Rating'] =
Quarterly_summary['Avg_Rating'].round(2)

Quarterly_summary
```

```
         Total_Sales  Avg_Rating  Total_Orders
Quarter
2025Q1    19664881.0        4.34         73085
2025Q2    19896936.0        4.34         74155
2025Q3    13441168.0        4.34         50163
```

1. Top 5 Cities by Sales

```
data.head(1)
```

```
       State       City Order Date  Month Year_Month Day_Name  \
0  Karnataka  Bengaluru 2025-06-29      6     2025-6   Sunday

            Restaurant Name              Location Category  \
0  Anand Sweets & Savouries  Rajarajeshwari Nagar    Snack

              Dish Name  Price (INR)  Rating  Rating Count
Food_Category  \
0  Butter Murukku-200gm        133.9     4.0             0
Veg

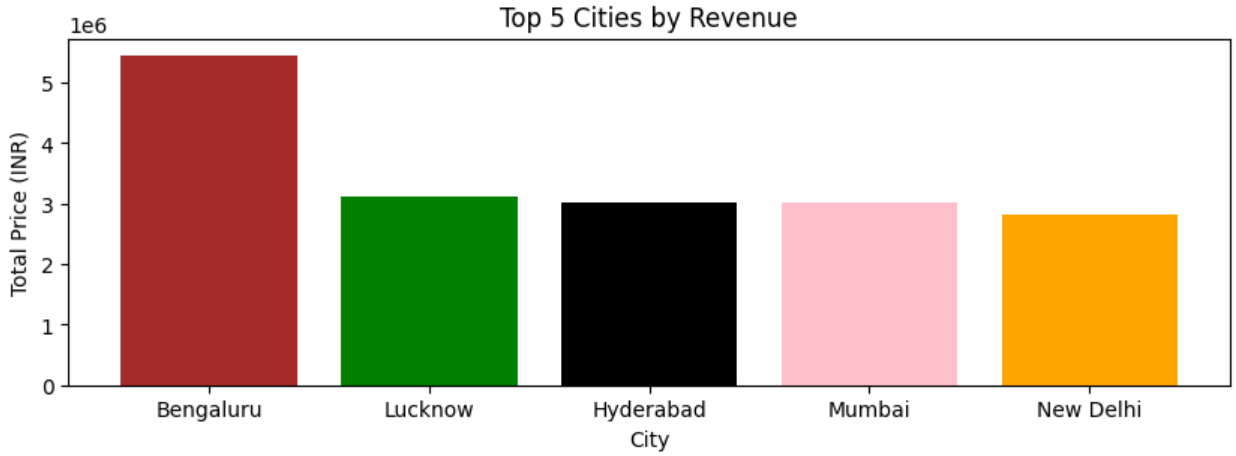
  Quarter
0  2025Q2
```

```
x = data.groupby('City')['Price (INR)'].sum().reset_index(name='Total
Price').sort_values('Total Price', ascending= False)[0:5]
x
```

```
         City  Total Price
3   Bengaluru   5455887.73
18    Lucknow   3117359.65
```

```
11  Hyderabad   3021711.62
19     Mumbai   3015573.35
20  New Delhi   2829180.60

plt.figure(figsize=(10,3))
plt.title('Top 5 Cities by Revenue')
plt.bar(x['City'], x['Total Price'], color =
['brown','green','black','pink','orange'])
plt.xlabel('City')
plt.ylabel('Total Price (INR)')
plt.show()
```



END